Reconstructing Historic Fossil Caves in X3D from Century-Old Survey Data: A Human–AI Case Study

Alexander Hoffman* Claude (Anthropic's Claude Code)

collaborative authorship; see Use of AI statement

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Abstract

We reconstruct two Pleistocene fossil caves of the McCloud River, California — *Potter Creek Cave* and *Samwel Cave*, both excavated in the early 1900s under the direction of the vertebrate palaeontologist John C. Merriam — as interactive, to-scale X3D scenes built procedurally from the original survey literature (Sinclair 1904; Furlong 1906; Feranec et al. 2007). As in our companion HAnim study, we treat the human–AI pair as co-authors: the human supplies intent, domain pointers, family knowledge, and aesthetic and ethical judgment; the AI supplies procedural generation, a tight verify-by-render loop, and contributions back to the toolchain. We report three results. First, a method for turning narrative cave surveys into standards-conformant X3D 4.0, organised around a *provenance discipline* that explicitly separates documented geometry from interpretive dressing. Second, a physically-based cave look-development pass — displaced and normal-mapped limestone (`PhysicalMaterial`), layered light-shaft effects, and rippled water — rendered in the X_ITE web player. Third, and most useful to the toolchain, an **X_ITE rendering backend** for the X3D Model Context Protocol (MCP) server that lets the tool *see* HAnim humanoids and X3D 4.0 `PhysicalMaterial` (physically-based rendering, PBR) scenes that the previous X3DOM-based path did not draw under our headless software-WebGL pipeline. A personal and ethical thread runs throughout: the excavator was the human author's great-great-grandfather, and the caves are Winnemem Wintu sacred places above a stretch of the McCloud River that Shasta Dam later flooded.

1 Introduction

This paper is a worked example of human–AI pairing applied to a problem that is equal parts engineering, archival scholarship, and cultural care. A human collaborator and an AI agent (Anthropic's Claude, in Claude Code) built two interactive X3D scenes reconstructing the chambers of *Potter Creek Cave* and *Samwel Cave* in Shasta County, California — to the dimensions recorded in their original surveys, and viewable in a web browser. The Model Context Protocol (MCP) [8] is an open protocol that exposes tools to a language model; the Web3D Consortium's X3D MCP server exposes X3D node lookup, scene construction, validation, and rendering as such tools.

The motivation was personal. The caves were excavated in the early 1900s by field parties under **John Campbell Merriam** (1869–1945), the University of California vertebrate palaeontologist who later led the Carnegie Institution and co-founded the Save the Redwoods League — and the human author's great-great-grandfather. Potter Creek Cave was excavated principally in 1902–1903

*Corresponding author: `alex@euphe.me`

and Samwel Cave in 1903–1906; the reports that resulted (Sinclair 1903 [11], 1904 [10], 1905 [16]; Furlong 1906 [12]) are among the founding documents of West Coast Pleistocene palaeontology. We asked: can a human–AI pair turn those century-old prose surveys into faithful, inspectable 3D — honestly labelled as to what is measured versus imagined — using only the open Web3D toolchain?

The contribution is less the scenes than the *method and the discipline*: every artifact was produced through the X3D MCP server, validated against the X3D schema and a set of semantic checks, and visually verified by the agent rendering each scene (Figs. 1–7) and reading the resulting image. Along the way we found and fixed the single largest gap in that loop — the MCP could not render the standard's current material model — and contributed the fix upstream.

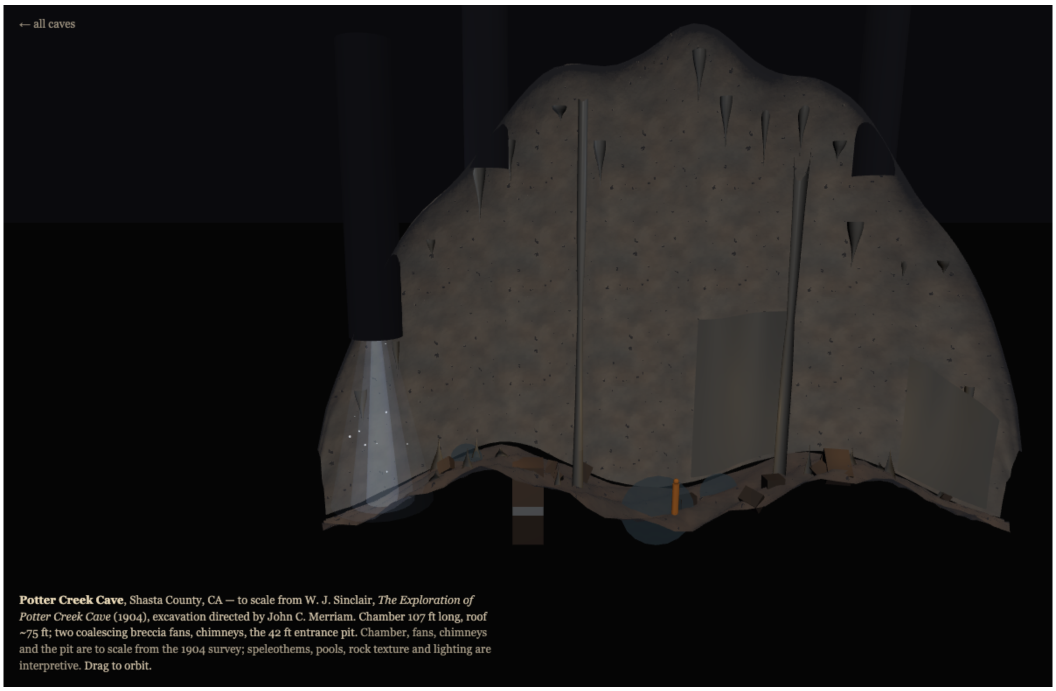

Figure 1: Potter Creek Cave, hero view near the entrance third, rendered in X_ITE. *Documented* to Sinclair (1904): the 107 ft chamber, ∼75 ft domed roof, the two coalescing breccia fans, and the 42 ft entrance pit (left, with its light shaft). *Interpretive*: speleothems, pools, rock texture, and lighting.

2 The sites and their sources

Potter Creek Cave. Sinclair's 1904 monograph [10] gives a chamber "one hundred and seven feet long, about thirty feet wide at its widest part, with the roof rising about seventy-five feet above the lowest point of the floor." Two "fan-like deposits of earth and stalagmite-cemented breccia" slope from opposite ends and coalesce in the middle, with near-vertical chimneys above each fan apex; access required "a vertical descent of forty-two feet" by rope ladder. The galleries trend northwest–southeast, parallel to the strike of the McCloud Limestone, at ∼1500 ft (460 m) elevation. The deposit yielded roughly 52 vertebrate species, 21 of them extinct — including the type specimen of the shrub-ox *Euceratherium collinum*, a new genus and species named from this cave (Sinclair & Furlong 1904) [13], abundant remains of the giant short-faced bear *Arctodus simus*, the Shasta ground sloth *Nothrotheriops shastensis*, *Megalonyx*, *Canis dirus*, mammoth, camel, and horse.

Polished bone and a stone chip recovered at depth raised an early human-association question, later evaluated by radiocarbon (Payen & Taylor 1976) [15].

Samwel Cave. Five kilometres north on the same McCloud arm, Samwel Cave (Wintu *sawal*) is a branching two-level system in the McCloud Limestone at ∼1500 ft (460 m) elevation (the scholarly literature spells it "Samwell"). *Sa-Wal* is the Wintu word for grizzly bear: the cave was holy because grizzly-bear spirits were held to dwell there, and Wintu men prayed there for strength before a hunt; other names and etymologies are also recorded. The cross-section in Feranec et al. (2007, Fig. 2) [14] names its spaces: the Entrance and Porcupine Entrance, the large Pleistocene Hall, the Gate, Chamber One, Merriam's Chamber, and Chamber Two ("Furlong's Room") — the lower level, reached only by a deep drop requiring climbing gear (Fig. 4). Furlong's 1906 report [12] itself carries no plan or section drawing — only photographs and an in-text stratigraphic column. Tradition records the cave as the *Cave of the Lost Maiden* — the Wintu girl Olchanolmet, who by legend fell to her death down a "90-foot-deep hole" — and as the *Cave of the Magic Pools*, where "Wintu medicine men bathed in the water pools to get magic strength." Nearly 1000 specimens (45 mammal, 13 bird species) span the Last Glacial Maximum (∼23,600–17,100 cal BC) [14].

An ethical note. Both caves are Winnemem Wintu sacred places. Shasta Dam (built 1935–45) flooded the McCloud River canyon — villages, burial and ceremonial places — and the caves now sit on the Shasta Lake shore. We name the Winnemem connection, the Lost Maiden, and the Magic Pools explicitly and with attribution, as context rather than scenery, and the dam-flooding is surfaced in each scene's on-screen caption.

3 Method: from narrative survey to standards-conformant X3D

Each cave is generated procedurally by a single Python program emitting X3D 4.0 XML, declared `profile='Immersive' version='4.0'` (ISO/IEC 19775-1:2023 [1]; XML encoding ISO/IEC 19776-1 [2]). Three ideas did most of the work.

Sectioning. A closed cave shell is a dark blob from outside and is unlit from within. We instead build the *back half* of each surface (the $z \leq 0$ portion), opening the chamber toward the viewer — the same longitudinal and cross-section conventions the 1904 and 1906 reports used to draw the caves on paper. This makes the interior simultaneously legible and lightable. Potter Creek becomes a lofted longitudinal section (Fig. 2); Samwel becomes a labelled cross-section of ellipsoidal chambers connected by passages, echoing Furlong's plate.

Physically based look-development. Walls and floor use the X3D 4.0 `PhysicalMaterial` node (Shape component [1]; its metallic-roughness model follows glTF 2.0 [7]) with a tileable texture set — base-colour, tangent-space normal, and metallic-roughness maps generated from value-noise — combined with multi-octave geometric displacement of the mesh for true silhouette relief. Note that `PhysicalMaterial` defaults `metallic=1` and `roughness=1` [1], so dielectric rock must explicitly set `metallic=0`. On top sit procedural speleothems (stalactites, stalagmites, fused columns, flowstone draperies), breakdown blocks, rippled PBR pools, and layered emissive light-shaft cones with drifting motes down the open shafts. The look was developed iteratively through the render loop (Fig. 3), each stage inspected before the next.

Figure 2: Potter Creek Cave as a lofted longitudinal section (back half, $z \leq 0$), to scale from Sinclair (1904): the domed roof, the two breccia fans, the chimney shafts, the banded stratigraphic column, and the light shaft down the 42 ft entrance pit. The back-half cut opens the interior so it both reads and lights.

Provenance discipline. Because the geometry is part measured and part imagined, every generator carries a `PROVENANCE` block, and every scene a caption, that each states plainly which features are *to scale from the survey* (chamber dimensions, the fans, the chimneys, the 42 ft pit; Samwel's chamber topology and the documented deep drop joining its two levels) and which are *interpretive* (all speleothems, the rock texture, pool extents, lighting, and — for Samwel — the individual room sizes, which the sources do not give). We are careful to separate documented fact from legend: that a deep drop joins Samwel's two levels is surveyed, whereas the maiden's "90-foot hole" is tradition, and our identification of the two is an interpretive choice. We regard this explicit separation as a small but reusable contribution to honest heritage and scientific reconstruction.

Documented drawings. At the opposite end of the provenance spectrum from the interpretive reconstruction, we also captured the survey drawings themselves. The original plates are line art, not text, so OCR (which gave us the chamber *names*) cannot recover their geometry; instead we extract the figure raster at full resolution and vectorise it (`pdftoppm` → crop → threshold → `potrace` → SVG). This reproduces each drawing identically as clean vector — Sinclair's (1904) topographic plan of the Potter Creek chamber floor (Pl. 14; Fig. 5), his longitudinal section (Pl. 11) and deposit cross-sections (Pls. 12–13), and the Samwel cross-section. A short converter (`svg_to_x3d.py`) then lifts the traced paths into an X3D `IndexedLineSet` to scale, so the documented plan becomes navigable geometry in the same web pipeline as the interpretive model — the *measured* extreme of the same provenance discipline.

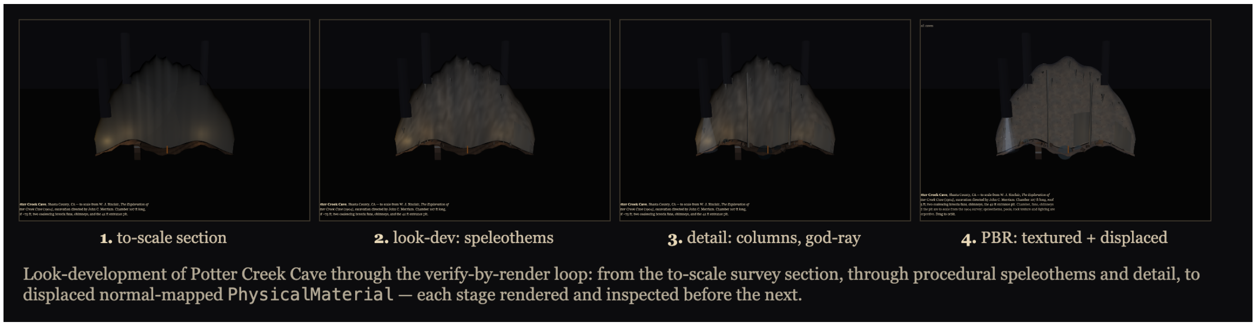

Figure 3: Look-development of Potter Creek Cave through the verify-by-render loop: from the to-scale survey section, through procedural speleothems and detail, to displaced normal-mapped `PhysicalMaterial`.

4 The render loop and the X3D MCP server

Our governing practice is *verify by render*: a scene that passes both the XSD schema check and our semantic checks can still be visually wrong — off camera, unlit, mis-scaled, or silently dropped. The canonical loop is therefore `describe_node` (look up exact fields and `containerFields`) → build → `validate_x3d` *and* `validate_semantic` → **render and look**. Both cave scenes pass both validators cleanly.

The decisive engineering finding was that the loop's last step was broken for exactly the scenes we care about. The MCP's `render_image` tool used X3DOM [9] with the synchronous Playwright API. Two failures compounded: (i) called from an asynchronous MCP tool it raised *"Sync API inside the asyncio loop"*; and (ii) under our headless pipeline (Chromium via Playwright 1.60, software WebGL via ANGLE/SwiftShader) X3DOM cleared the background but drew no geometry for either scene. We state this as the empirical result we observed, scoped to what we tested — X3DOM from `x3dom.org/release` (1.8.x) under headless SwiftShader — a rendering-coverage and headless-software-WebGL gap, not a blanket statement about X3DOM in all configurations. (X3DOM does not support HAnim [3]; its X3D 4.0 `PhysicalMaterial` coverage is limited.) The controlled result is shown in Fig. 6: two identical scene files — the PBR cave and the HAnim slalom-runner of our companion study — render empty under X3DOM and fully under X_ITE, which is the capability (PBR *and* HAnim) the Web3D community most needs an authoring agent to be able to see.

We rebuilt the backend around **X_ITE** [5], which renders both PBR and HAnim. The new implementation (i) uses asynchronous Playwright so it runs inside the MCP event loop; (ii) serves the scene over an ephemeral loopback HTTP server, because X_ITE fetches the `.x3d` by XHR and the browser's same-origin policy blocks that for `file://` URLs (reported as a CORS error); and (iii) when given a file path, serves that file's own directory so relative assets (PBR textures, inlined geometry) resolve as they do in a browser. With this in place the agent could finally inspect its own output; every rendered figure here was produced through it, pinned to X_ITE 15.1.4 for reproducibility.

5 Findings

1. **Renderer choice is a correctness property, not an aesthetic one.** Under our headless setup X3DOM rendered neither the PBR cave nor the HAnim scene, whereas X_ITE rendered

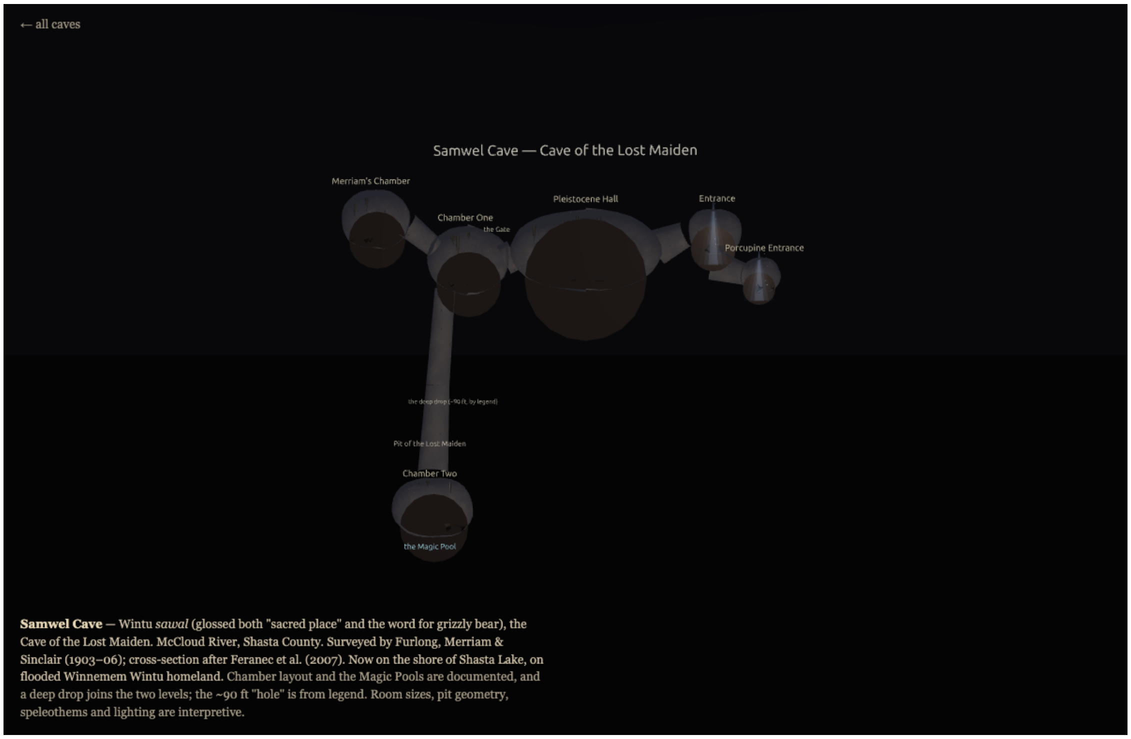

Figure 4: Samwel Cave as a labelled cross-section after Feranec et al. (2007, Fig. 2). *Documented* topology: Entrance and Porcupine Entrance, Pleistocene Hall, the Gate, Chamber One, Merriam's Chamber, and Chamber Two down a deep drop to the lower level with its Magic Pool. Room sizes and surface detail are interpretive.

both. An authoring tool that cannot render the standard's current material model cannot verify its own work.

2. **Transport matters.** `file://` silently defeats X_ITE (the same-origin policy for local files); inline X3D inside HTML is mangled by the HTML parser's tag lowercasing. A small local HTTP server is the reliable path, and pages should *detect* `file://` and say so rather than show a black screen.

3. **Look up, do not recall.** Every material field (`baseTexture`, `normalTexture`, `metallicRoughnessTexture`; `metallic`, default 1, set to 0 for rock) came from `describe_node`, not memory. Guessing `containerField`s is the most common silent failure, and the semantic checks that name the exact fix are worth more than the schema check alone.

4. **Provenance is cheap and clarifying.** Recording what is measured versus imagined took minutes, removed a class of future mis-citation, and forced better research — it is how we caught that the maiden's 90-foot fall is *legend* while the level-joining drop is what the *survey* attests, and kept the two distinct.

5. **Get the legend right.** A late instruction — "read the map legends correctly" — sent us back to the published cross-section to verify chamber identities and to surface the Magic Pool, which then became the emotional and geometric focus of the Samwel descent view (Fig. 7).

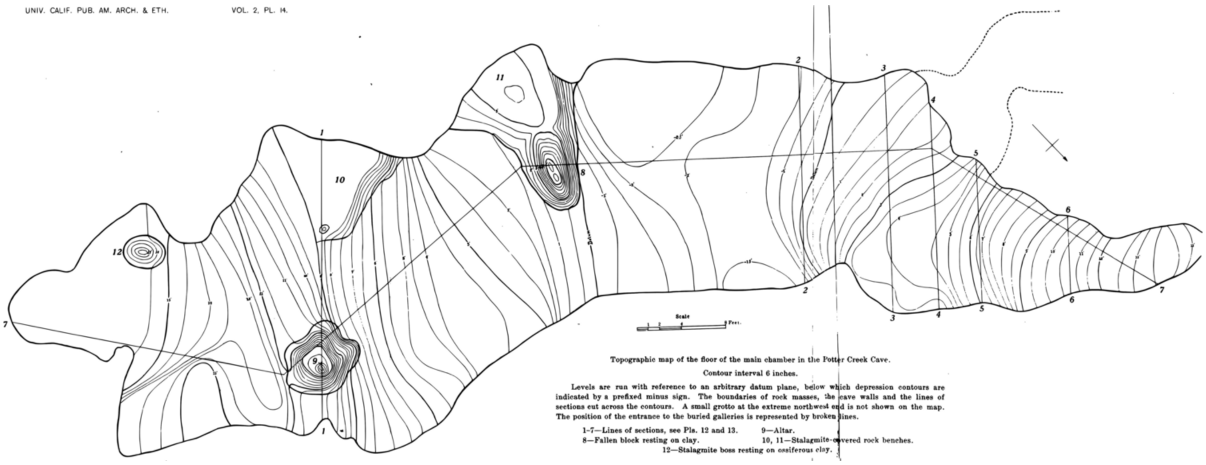

Figure 5: The *documented* extreme: a faithful vector trace of Sinclair's (1904, Plate 14) "topographic map of the floor of the main chamber in the Potter Creek Cave, contour interval 6 inches" — floor contours, section lines 1–7, stalagmite bosses, the altar, and the buried galleries, reproduced identically and lifted to a to-scale X3D `IndexedLineSet`.

6 Ideas and future work

- **Populate the caves with their fauna.** The procedural-creature pipeline already built for a rigged moose could place the actually-excavated megafauna — the type *Euceratherium*, the short-faced bear, the Shasta ground sloth — at their recorded depths, turning each cave into a labelled palaeontological diorama.
- **Ground-truth from the field.** An in-person visit to Samwel could supply photographs of the real chambers and formations, replacing estimated room sizes and invented dripstone with measured shapes — closing the interpretive gap the provenance block currently flags.
- **Stratigraphic interactivity.** Expose the documented strata (Potter Creek's clay/ash/breccia; Samwel's flowstone–breccia–gravel column) as a scrubbable cut, with fauna pinned to levels and radiocarbon ages on hover.
- **Toward true volumetrics.** Our light shafts and "refractive" water are faked with layered translucent geometry; X3D's `Fog`/`LocalFog` give distance attenuation but no participating-media *in-scattering* primitive for god-ray-style effects (the VolumeRendering component targets sampled volume data, not atmosphere). A screen-space or volumetric approach in X_ITE would be a genuine rendering contribution.
- **A reusable survey→X3D recipe.** The section-cut + PBR + provenance pattern generalises to any cave or excavation with a published plan; it could become a small documented template for the AI-with-X3D community.

7 Contributions back to the toolchain

Beyond the scenes, the work produced concrete improvements to the Web3D X3D MCP server, carried on a dedicated branch and covered by tests: the X_ITE rendering backend described above (async, loopback-served, asset-aware); semantic validators (built from the X3D Unified Object Model, X3DUOM [4]) for `containerField` (with the canonical field suggested), USE-before-DEF

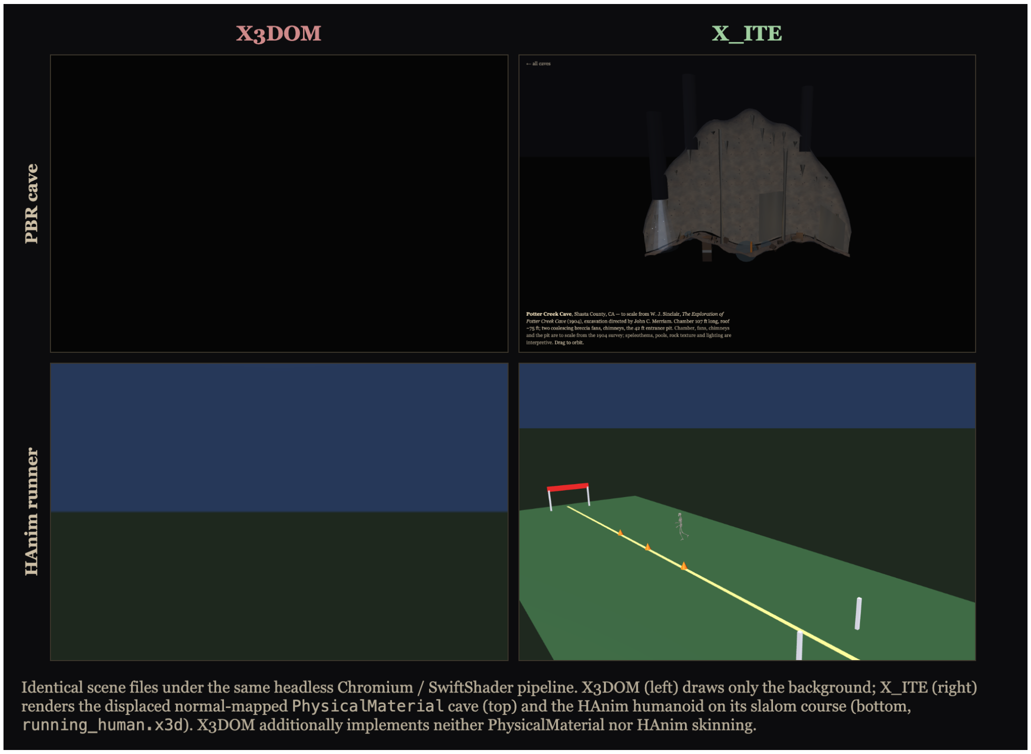

Figure 6: Controlled comparison: identical scene files under the same headless Chromium/Swift-Shader pipeline. X3DOM (left) draws only the scene background; X_ITE (right) renders the displaced, normal-mapped `PhysicalMaterial` cave (top) and the HAnim humanoid on its slalom course (bottom, `running_human.x3d`). Beyond this headless result, X3DOM does not support HAnim and has limited X3D 4.0 `PhysicalMaterial` coverage. This motivated the new X_ITE render backend.

ordering, and interpolator `key`/`keyValue` length; an `autofix_x3d` tool that rewrites `containerFields` and returns the corrected document; granular-mode parity for non-default node placement; and proactive server `instructions` that frame the canonical describe → build → validate-both → render path for fresh model instances. We also maintain an upstream report against **x3d.py** [6], the Web3D Consortium's official X3D Python (Scene Access Interface) package, version 4.0.65.3: it drops `containerField` on non-default fields and defaults `EnvironmentLight.global` to `true` where the normative spec specifies `false` — both of which serialise to schema-valid XML that nonetheless renders incorrectly.

8 Conclusion

Two caves that a family forebear opened a hundred and twenty years ago now exist as honest, inspectable, standards-conformant 3D, made entirely through the open Web3D toolchain by a human and a model working as a pair. The engineering lesson is blunt: an authoring agent is only as good as its ability to see what it made, and for X3D 4.0 material that means an X3D-4.0-complete renderer such as X_ITE. The human lessons — provenance, attribution, and care for whose sacred ground this is — were supplied by the person, and they are the part that matters most.

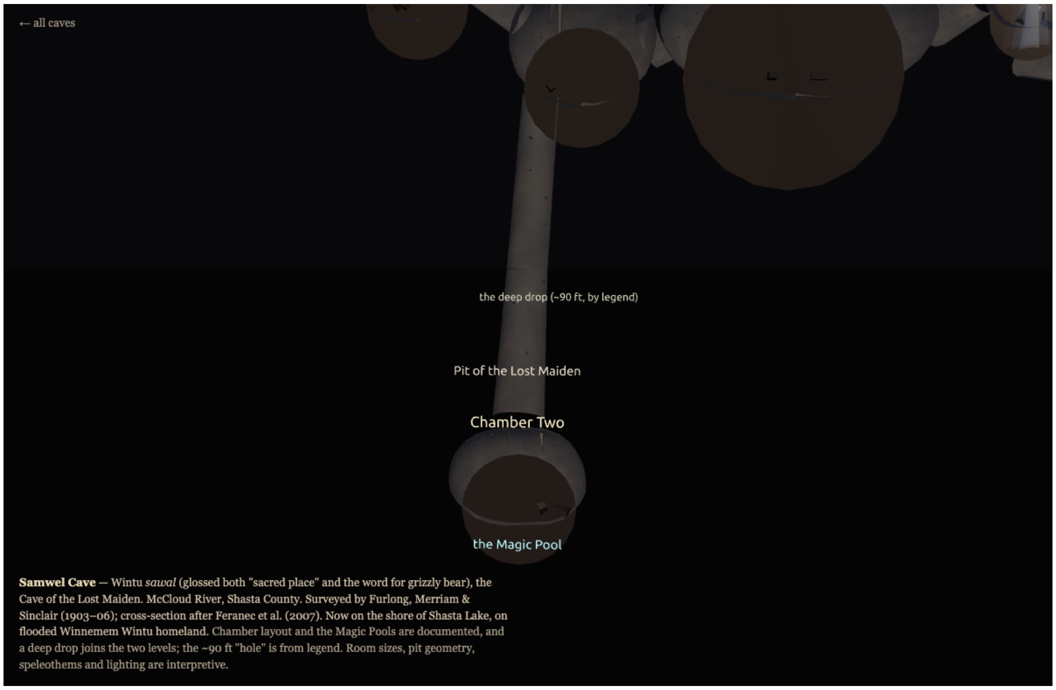

Figure 7: Samwel Cave, the descent: Chamber One, the deep drop (the ∼90 ft "hole" of legend), and Chamber Two with the glowing Magic Pool at the bottom. The on-screen caption marks documented vs. interpretive features and the contested gloss of *sawal*.

Use of AI. The named AI "co-author" is Anthropic's Claude (Claude Code), which performed procedural generation, rendering, validation, and drafting under the human author's direction; the human author conceived the project, supplied sources and family knowledge, made all aesthetic and ethical decisions, and is responsible for the content. We list the AI on the byline for transparency; venues that bar AI authorship should treat this as a tool-use disclosure.

Reproducibility. All generators, textures, scenes, viewer pages, and the full render history are in the project repository (branch `potter-creek-cave`); `./start_caves.sh` serves the launcher locally. Renders used X_ITE 15.1.4, X3DOM `x3dom.org/release` (1.8.x), Playwright 1.60 with headless Chromium and ANGLE/SwiftShader software WebGL. The exact commit/tag accompanies publication.

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